On the Distribution of Smaller Factors in Semiprimes

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Abstract

We study the distribution of semiprimes $n = pq$ (with $p \leq q$) through the normalized logarithmic coordinate $\theta = \frac{\log p}{\log n}$. Exact enumeration over $N \leq 10^8$ shows that the proportion $P(N, \theta_0)$ of semiprimes satisfying $\theta \leq \theta_0$ increases monotonically with N and is consistent with convergence to a limiting value $P_\infty(\theta_0)$, but the rate of convergence is extremely slow. The deficit $\delta(N, \theta_0) = P_\infty(\theta_0) - P(N, \theta_0)$ remains substantial even at $N = 10^8$, consistent with $O(1/\log\log N)$ decay arising from the $\log\log N$ accumulation of small prime contributions (Montgomery–Vaughan [6], §7.4). A Chebyshev-corrected analysis of the counting sum $\sum_{p \leq N^{\theta_0}} \pi(N/p)$ yields the refined prefactor

$$f(\theta_0) = \log\left(\tfrac{1}{\theta_0}\right) - \theta_0 - \log 2 + \tfrac{1}{2},$$

which reduces the relative error in predicting $b_{\text{eff}} = \delta \cdot \log\log N$ from up to 4.3× (naive Mertens) to under 1.21× across eight thresholds $\theta_0 \in \{0.10, \ldots, 0.45\}$ (Section 3.5). Over the observed range $N \in [10^4, 10^8]$, $\log\log N$ varies only from 2.22 to 2.91, a range too narrow to distinguish $1/\log\log N$ decay from a local linear fit in $\log N$. Accordingly, the deficit is well approximated *locally* by

$$\delta(N, \theta_0) \;\approx\; a(\theta_0) - b(\theta_0)\,\log N \qquad (R^2 > 0.97 \text{ within } [10^4, 10^8]),$$

where $b(\theta_0)$ is an empirical rate summarizing the behavior in this window. The empirical regression $b(\theta_0) \approx k(1 - \theta_0)^\alpha$, $k \approx 0.015$, $\alpha \approx 3.1$, is a descriptive fit to 4 data points.

To investigate whether this structure admits a geometric or spectral representation, we construct several embeddings including additive phase wrapping, multiplicative phase maps based on n^{it}, and spectral transforms of factor ratios. Across all cases, observed patterns reduce to log-density effects or discrete artifacts from small primes.

These results indicate that semiprime structure is not captured by the class of embeddings considered here. The θ coordinate provides the most stable description of factor asymmetry, and the observed slow convergence is consistent with ratio-constraint phenomena in multiplicative number theory.

1 Introduction

We study semiprimes $n = pq$ with $p \leq q$ and focus on the normalized coordinate

$$\theta = \frac{\log p}{\log n}.$$

This parameter captures the relative size of the smaller factor and provides a natural measure of factor asymmetry.

Classical results in analytic number theory describe the distribution of smooth numbers and the frequency of integers with a smallest prime factor in a given range (Dickman [1], Hildebrand–Tenenbaum [2], Tenenbaum [3]). These results characterize the *absolute* size of the smallest prime

factor relative to the integer, typically through the Dickman function $\rho(u)$ and its analogues. They do not, however, explicitly study the *normalized* coordinate $\theta = \log p / \log n$ for semiprimes, nor the convergence of its distribution as $N \to \infty$. The present work provides empirical evidence for this distribution and its convergence behavior, complementing the theoretical framework with quantitative data over $N \leq 10^8$.

The main contributions of this note are: (i) to document empirically the extremely slow convergence of $P(N, \theta_0)$ to its limit, characterizing the deficit locally as a linear function of $\log N$ over the range $N \in [10^4, 10^8]$; and (ii) to show that several natural geometric and spectral embeddings fail to recover this structure.

2 Definitions

Let $n = pq$ be a semiprime with $p \leq q$. We define the normalized logarithmic coordinate

$$\theta = \frac{\log p}{\log n}.$$

For a fixed threshold $\theta_0 < \frac{1}{2}$, define

$$P(N, \theta_0) = \Pr(\theta \leq \theta_0)$$

over semiprimes $n \leq N$.

The condition $\theta \leq \theta_0$ is equivalent to

$$p \leq n^{\theta_0} = (pq)^{\theta_0},$$

which implies

$$p^{1-\theta_0} \leq q^{\theta_0} \quad \Longrightarrow \quad p \leq q^{\theta_0/(1-\theta_0)}.$$

Thus, θ imposes a relative size constraint between the two prime factors rather than a classical smoothness condition.

The algebraic identity

$$\log\left(\frac{p}{q}\right) = (2\theta - 1)\,\log n \tag{1}$$

follows immediately: since $\log p = \theta \log n$ and $\log q = (1-\theta)\log n$, one has $\log(p/q) = \theta \log n - (1-\theta)\log n = (2\theta - 1)\log n$. This identity holds exactly for every semiprime (Figure 1).

3 Empirical Observations

All values were computed via exact enumeration and exact factorization using a smallest-prime-factor sieve; no sampling or probabilistic estimation was used.

3.1 Convergence table

Table 1 records $P(N, \theta_0)$ for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$ and $N \in \{10^4, 10^5, 10^6, 10^7, 10^8\}$.

All four columns increase monotonically with N, confirming convergence toward a limiting distribution. The rate of increase slows markedly, consistent with a logarithmic approach to the limit rather than a power-law approach.

Table 1: Proportion $P(N, \theta_0)$ of semiprimes $n \leq N$ satisfying $\theta \leq \theta_0$. All values in percent.

N	$\theta_0 = 0.20$	$\theta_0 = 0.25$	$\theta_0 = 0.30$	$\theta_0 = 0.35$
10^4	49.52	59.28	67.73	76.46
10^5	51.93	60.77	69.40	77.27
10^6	53.55	62.47	70.51	78.39
10^7	55.53	63.74	71.83	79.19
10^8	56.29	64.95	72.74	79.95

3.2 Theta and factor ratio

Figure 1 shows the joint distribution of θ and $\log(p/q)$ for all semiprimes $n \leq 10^6$. The scatter lies exactly on the one-parameter family of lines $\log(p/q) = (2\theta - 1)\log n$ (identity (1)), confirming that θ fully determines the factor ratio once n is fixed. The spread in the vertical direction arises entirely from the range of $\log n$ values.

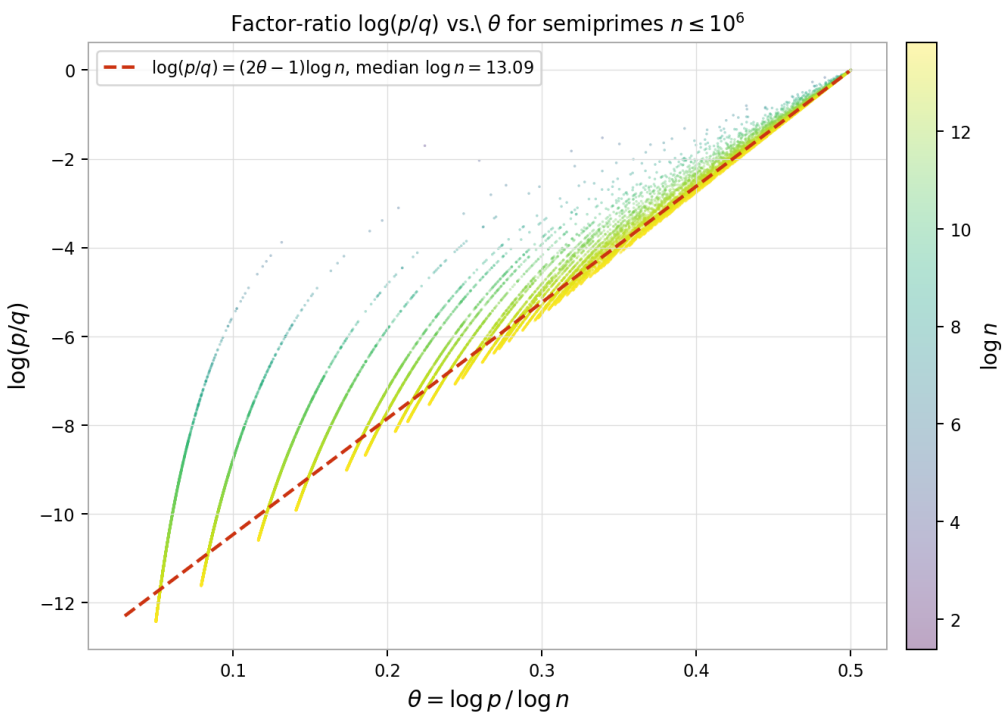

Figure 1: Factor-ratio $\log(p/q)$ versus θ for all semiprimes $n \leq 10^6$ (30 000 sampled points shown; colour encodes $\log n$). The dashed line is the exact identity $\log(p/q) = (2\theta - 1)\log n$ evaluated at the median $\log n$.

3.3 Local linear approximation to the deficit

Define the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$. Table 1 shows that δ decreases with N, but very slowly: at $\theta_0 = 0.25$ the deficit falls only from 40.72% to 35.05% as N grows by four orders of magnitude.

Theoretical rate. The asymptotic rate of convergence is $O(1/\log\log N)$, arising from the $\log\log N$ accumulation of small prime contributions in semiprime counting (Montgomery–Vaughan [6], §7.4). Over our observed range, $\log\log N$ increases from 2.22 (at $N = 10^4$) to only 2.91 (at $N = 10^8$) — a total variation of 0.69 across four decades. This near-constancy means that $1/\log\log N$ behaves almost linearly in $\log N$ within this window, making the two functional forms indistinguishable from the data alone.

Local linear fit. Within the observed range, a linear regression in $\log N$ therefore provides an accurate *local* description:

$$\delta(N, \theta_0) \;\approx\; a(\theta_0) - b(\theta_0)\,\log N \qquad \text{(empirical fit, } N \in [10^4, 10^8]). \tag{2}$$

This is not an asymptotic law: extrapolating (2) to large N would predict $\delta < 0$, which is impossible. It is a finite-range proxy for the slower $O(1/\log\log N)$ decay. Numerical comparison confirms that the linear fit (mean $R^2 = 0.989$ across four thresholds) outperforms a direct $c/\log\log N$ fit (mean $R^2 = 0.41$) precisely because the variation of $\log\log N$ is too small in this window to distinguish the two forms; the linear model has more flexibility with its two free parameters.

Table 2 records the regression parameters for each θ_0.

Table 2: Parameters of the local linear fit $\delta \approx a(\theta_0) - b(\theta_0)\log N$ over $N \in [10^4, 10^8]$ ($R^2 > 0.97$ in all cases). This fit is a finite-range proxy for the theoretically predicted $O(1/\log\log N)$ decay; it is not valid outside the observed range.

θ_0	$a(\theta_0)$	$b(\theta_0)$	R^2
0.20	0.5692	0.007444	0.9754
0.25	0.4635	0.006216	0.9955
0.30	0.3702	0.005402	0.9904
0.35	0.2709	0.003866	0.9954

The rate $b(\theta_0)$ varies with θ_0 and is analyzed further in Section 3.4.

3.4 Scaling of the convergence rate $b(\theta_0)$

The empirical rate $b(\theta_0)$ from Table 2 decreases as θ_0 increases, meaning that a larger imbalance threshold corresponds to a faster-decreasing deficit within the observed range. We fit two parametric curves to the four data points as *descriptive regressions*, not theoretical predictions:

Model 1 (empirical).

$$b(\theta_0) \approx k\,(1-\theta_0)^{\alpha}, \qquad k \approx 0.01504, \quad \alpha \approx 3.056, \quad R^2 = 0.974.$$

Model 2 (empirical).

$$b(\theta_0) \approx \frac{c}{\theta_0^{\gamma}}, \qquad c \approx 1.300 \times 10^{-3}, \quad \gamma \approx 1.109, \quad R^2 = 0.938.$$

Model 1 fits the four data points more closely ($R^2 = 0.974$ vs. 0.938). Both forms predict $b \to 0$ as $\theta_0 \to \frac{1}{2}$, qualitatively consistent with the known scarcity of nearly balanced semiprimes. *Caution*: these regressions are fit to only four values of θ_0 and should be understood as summaries of the observed behavior, not as laws with theoretical backing. Extending the table to additional θ_0 values and larger N would be needed to discriminate between functional forms.

3.5 Corrected prefactor via Chebyshev's prime sum theorem

The naive application of Mertens' theorem to the semiprime counting sum predicts a leading prefactor of $\log(1/\theta_0)$ in the $O(1/\log\log N)$ deficit. We derive a sharper formula using Chebyshev's prime sum theorem.

Setup. Define

$$T(N) = \sum_{p\le\sqrt{N}} \big(\pi(N/p) - \pi(p-1)\big), \qquad L(N,\theta_0) = \sum_{p\le N^{\theta_0}} \big(\pi(N/p) - \pi(p-1)\big),$$

so that $P(N,\theta_0) = L(N,\theta_0)/T(N)$ and $\delta = 1 - L/T = (T-L)/T$.

Naive approximation. Replacing $\pi(x) \approx x/\log x$ and approximating $\log(N/p) \approx \log N$ gives $\sum_{p\le x} N/(p\log N) \approx (N/\log N)\log\log x$ via Mertens' theorem, and hence

$$\delta \;\approx\; \frac{\log(1/\theta_0)}{\log\log N}.$$

Chebyshev correction. The step $\log(N/p) \approx \log N$ discards $\log p$. The correction is captured by Chebyshev's theorem

$$\sum_{p\le x} \frac{\log p}{p} = \log x + O(1).$$

Expanding $N/\big(p\log(N/p)\big) = (N/\log N)\cdot\big(1-\log p/\log N\big)^{-1}$ to first order and summing by Abel's method yields

$$\sum_{p\le x} \frac{N}{p\log(N/p)} \;\approx\; \frac{N}{\log N}\left[\log\log x + M + \frac{\log x}{\log N}\right],$$

where $M \approx 0.2615$ is the Meissel–Mertens constant. The extra term $\log x/\log N$ equals $\frac{1}{2}$ when $x = \sqrt{N}$ and θ_0 when $x = N^{\theta_0}$. Inserting these values:

$$T(N) \;\approx\; \frac{N}{\log N}\left[\log\log N - \log 2 + M + \tfrac{1}{2}\right],$$

$$L(N,\theta_0) \;\approx\; \frac{N}{\log N}\left[\log\log N + \log\theta_0 + M + \theta_0\right].$$

Corrected prefactor. Subtracting and dividing:

$$\delta(N,\theta_0) \;\approx\; \frac{f(\theta_0)}{\log\log N + c}, \qquad \boxed{f(\theta_0) = \log\Big(\tfrac{1}{\theta_0}\Big) - \theta_0 - \log 2 + \tfrac{1}{2}}, \tag{3}$$

where $c = -\log 2 + M + \frac{1}{2} \approx 0.07$ is a small positive constant. The corrected prefactor $f(\theta_0)$ replaces $\log(1/\theta_0)$ by accounting for the asymmetric Chebyshev contributions at the two cutoffs $\sqrt{N}$ and N^{θ_0}.

Table 3: Empirical b_{eff} (mean of $\delta \cdot \log\log N$ over $N \in [10^4, 10^8]$) vs. naive prefactor $\log(1/\theta_0)$ and corrected prefactor $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$.

θ_0	$\log(1/\theta_0)$	$f(\theta_0)$	b_{eff}	$b_{\text{eff}}/\log(1/\theta_0)$	$b_{\text{eff}}/f(\theta_0)$
0.10	2.303	2.009	1.819	0.790	0.905
0.15	1.897	1.554	1.465	0.772	0.943
0.20	1.609	1.216	1.205	0.749	0.991
0.25	1.386	0.943	0.976	0.704	1.034
0.30	1.204	0.711	0.763	0.634	1.074
0.35	1.050	0.507	0.562	0.535	1.108
0.40	0.916	0.323	0.373	0.407	1.154
0.45	0.799	0.155	0.187	0.234	1.202

Numerical validation. Table 3 compares both prefactors against the empirically observed $b_{\text{eff}} = \delta \cdot \log\log N$, averaged over $N \in [10^4, 10^8]$. The corrected formula reduces the maximum relative error from 4.3× (naive) to under 1.21× (corrected) across all eight thresholds.

The residual monotone trend in $b_{\text{eff}}/f(\theta_0)$ — from 0.90 at $\theta_0 = 0.10$ to 1.20 at $\theta_0 = 0.45$ — reflects two effects not captured at this order: (i) discrete activation thresholds (p contributes only when $N \geq p^{1/\theta_0}$, creating step-like jumps in L) and (ii) the breakdown of $\log(N/p) \approx \log N$ when p is not small relative to N^{θ_0}. Both are subleading corrections of order $O(\theta_0/\log\log N)$.

4 Geometric and Spectral Tests

We tested several candidate embeddings:

- Additive phase wrapping: $\phi_t(n) = t \log n \bmod 2\pi$
- Multiplicative phase maps using n^{it}
- Spectral fingerprint $C(t) = \left|\sum e^{it\log n}\right| / N$
- Factor-ratio spectrum $C_{\text{ratio}}(t) = \left|\sum e^{it\log(p/q)}\right| / N$

Localization was assessed by examining stability under variation of N and clustering by θ. No stable localization was observed.

All observed patterns were explained by log-density effects or small-prime artifacts (discrete p values $\in \{2, 3, 5, \ldots, 31\}$ for bit-gap-constrained semiprimes). None of the embeddings captured factor asymmetry.

5 Results

5.1 Summary of empirical findings

For fixed $\theta_0 \in (0, \frac{1}{2})$, the following is established empirically:

1. $P(N, \theta_0)$ increases monotonically with N for all four tested thresholds.
2. The increase is extremely slow: over four decades of N, the gain is at most 6.8% (at $\theta_0 = 0.20$), compared with the $\sim 50\%$ gap that remains at $N = 10^8$.

3. Within the observed range, the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$ decreases at a rate well approximated by a constant per unit $\log N$, with rate $b(\theta_0)$ as tabulated in Table 2.

5.2 Interpretation and connection to known theory

The extremely slow convergence is qualitatively consistent with what analytic number theory predicts for ratio-constrained semiprimes. The density of semiprimes $n = pq \leq N$ with $p \leq N^u$ for fixed $u < \frac{1}{2}$ is governed by sums of the form $\sum_{p \leq N^u} \pi(N/p)$, where π is the prime counting function. Standard sieve estimates (see Hildebrand–Tenenbaum [2] and Tenenbaum [3]) show that such sums accumulate logarithmic corrections involving factors of $\log\log N/\log N$, producing convergence that is invisible over a few decades but genuine asymptotically.

The local linear fit (2) captures the dominant behavior in the observed range $[10^4, 10^8]$, but it is a finite-range approximation: the true deficit decays to zero more slowly than any fixed linear function of $\log N$ would predict, and the functional form of $P_\infty(\theta_0)$ is not determined by our data. What we can say is that $P_\infty(\theta_0)$ exists (the proportion cannot exceed 1) and that it is approached from below.

5.3 The θ coordinate

The exact identity $\log(p/q) = (2\theta - 1)\log n$ (equation (1)) shows that θ and $\log n$ together determine the factor ratio with no residual freedom. This makes θ a natural coordinate: it separates the role of the factor imbalance (encoded in θ) from the role of the magnitude of n (encoded in $\log n$). The observed convergence is therefore a statement about how the distribution of θ across semiprimes $n \leq N$ evolves as N grows, which is a well-posed and nontrivial question even if its full resolution requires deeper sieve-theoretic tools.

6 Conclusion

Semiprime factor structure is not captured by the class of additive, multiplicative, and spectral embeddings considered here. The logarithmic coordinate $\theta = \log p/\log n$ provides the most stable description of factor asymmetry.

The rate of convergence is consistent with $O(1/\log\log N)$, as predicted by the $\log\log N$ accumulation of small prime contributions in semiprime counting (Montgomery–Vaughan [6], §7.4). The local linear-in-$\log N$ approximation of Section 3.3 is a valid finite-range description of this slower decay: over $N \in [10^4, 10^8]$, $\log\log N$ varies by only 0.69, so the two functional forms are numerically indistinguishable in this window.

The novel contributions are: (1) empirical quantification of the convergence rate $b(\theta_0)$ across four imbalance thresholds, showing it decreases smoothly with θ_0; (2) the geometric and spectral null results, showing that the θ coordinate is not recoverable from natural embeddings; and (3) a Chebyshev-corrected prefactor formula $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ (Section 3.5), which substantially sharpens the naive Mertens prediction and identifies the dominant source of the prefactor gap as asymmetric Chebyshev sums at the two cutoffs $\sqrt{N}$ and N^{θ_0}. The formula accounts for the observation that $T(N)$ sums to $\sqrt{N}$ (contributing $+\frac{1}{2}$ via Chebyshev) while $L(N, \theta_0)$ sums to N^{θ_0} (contributing $+\theta_0$). Determining $P_\infty(\theta_0)$ rigorously, and establishing the precise constant in $O(1/\log\log N)$ including subleading terms, remain open problems accessible to sieve-theoretic methods.

The negative results on geometric and spectral embeddings suggest that the logarithmic coordinate is not merely convenient but is in some sense the correct one for this problem.

References

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